

Discrete Root

The problem of finding a discrete root is defined as follows. Given a prime n and two integers a and k , find all x for which:

$$x^k \equiv a \pmod{n}$$

The algorithm

We will solve this problem by reducing it to the [discrete logarithm problem](#).

Let's apply the concept of a [primitive root](#) modulo n . Let g be a primitive root modulo n . Note that since n is prime, it must exist, and it can be found in $O(\text{Ans} \cdot \log \phi(n) \cdot \log n) = O(\text{Ans} \cdot \log^2 n)$ plus time of factoring $\phi(n)$.

We can easily discard the case where $a = 0$. In this case, obviously there is only one answer: $x = 0$.

Since we know that n is a prime and any number between 1 and $n - 1$ can be represented as a power of the primitive root, we can represent the discrete root problem as follows:

$$(g^y)^k \equiv a \pmod{n}$$

where

$$x \equiv g^y \pmod{n}$$

This, in turn, can be rewritten as

$$(g^k)^y \equiv a \pmod{n}$$

Now we have one unknown y , which is a discrete logarithm problem. The solution can be found using Shanks' baby-step giant-step algorithm in $O(\sqrt{n} \log n)$ (or we can verify that there are no solutions).

Having found one solution y_0 , one of solutions of discrete root problem will be $x_0 = g^{y_0} \pmod{n}$.